

K2-30b

R-2450

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_201027 · created 2026-08-22T23:58:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459149.9426 +/- 0.0083 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.047
Transit depth (Rp/Rs)^2	1.89 +/- 1.29 [%]
Orbital Inclination (inc)	86.04 +/- 2.33
Airmass coefficient 1 (a1)	1.257 +/- 0.035
Airmass coefficient 2 (a2)	-0.155 +/- 0.041
Scatter in the residuals of the lightcurve fit is	2.81 %
Best Comparison Star	None
Optimal Aperture	3.66
Optimal Annulus	12.43
Transit Duration (day)	0.058 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.047 vs 0.1266 (deviation 0.22 σ)
Duration fit vs published	0.058 d vs 0.0971 d (deviation 1.03 σ)
Mid-time O-C	15.2 min
Depth SNR	2.8
Residual scatter	2.81 %

Light curve

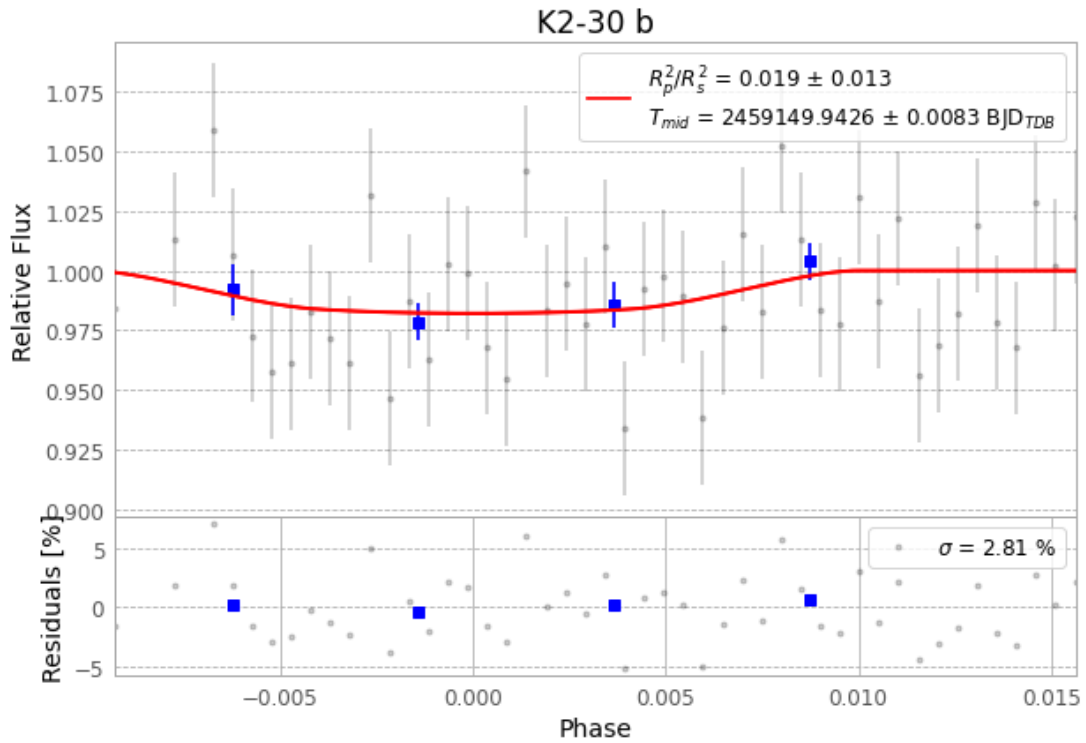


Figure 1 — FinalLightCurve_K2-30 b_2020-10-27.png (SHA-256 3780b0c2a17855c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 309], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 59de115f7a580b56...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), K2-30201027093012.FITS → K2-30201027120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-201027.log (11458a26c498...), FinalLightCurve_K2-30 b_2020-10-27.png (3780b0c2a178...), FinalLightCurve_K2-30 b_2020-10-27.csv (02f4285a88a3...)

Compute resources

Wall time 123.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:58Z.